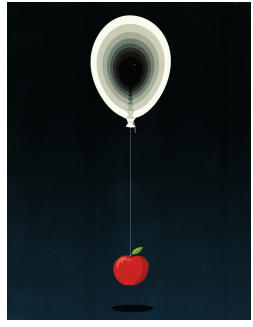
Should gravity be quantized?

my journey in foundations



Newscientist

Antoine Tilloy

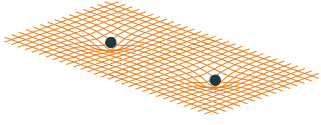
Theory division workshop

January 18th, 2021

MPQ

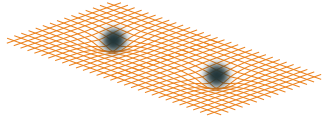
The biggest question in fundamental physics

Classical gravity



- **Matter** is classical
- Spacetime is classical

Semiclassical gravity



- **Matter** is quantum
- Spacetime is classical

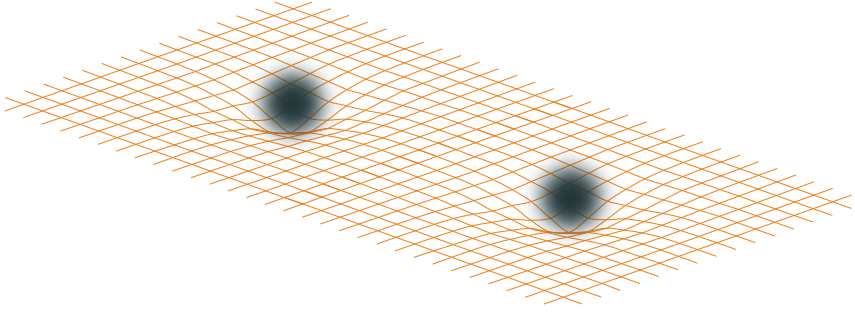
Fully quantum gravity



- **Matter** is quantum
- Spacetime is quantum

The last two are in principle possible and compatible with what we know

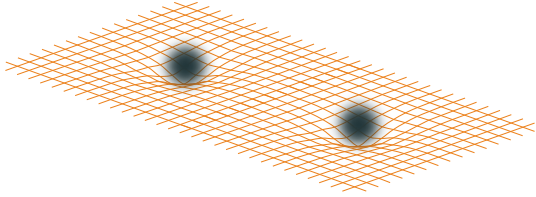
Standard semiclassical gravity



“Standard” semi-classical gravity

A semi-classical theory of gravity tells 2 stories:

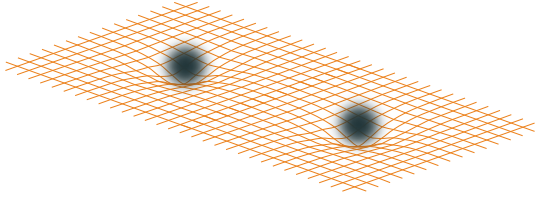
1. Quantum matter moves in a curved classical space-time
2. The classical space time is curved by quantum matter



“Standard” semi-classical gravity

A semi-classical theory of gravity tells 2 stories:

1. Quantum matter moves in a curved classical space-time
2. The classical space time is curved by quantum matter



1 is known (QFTCST), **2** is not

The crucial question of semi-classical gravity is to know how quantum matter should source curvature.

Møller-Rosenfeld semi-classical gravity

The **CHOICE** of Møller and Rosenfeld it to take:

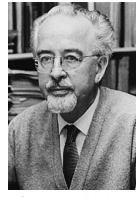
$$R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu} = 8\pi G \langle \hat{T}_{\mu\nu} \rangle$$

$\rightarrow$ source gravity via expectation values

There are:

- **technical relativistic** difficulties
  [renormalization of $\langle T_{\mu\nu} \rangle$, cosmological constant]
- **conceptual non-relativistic** difficulties
  [Born rule, faster-than-light signaling $\cdots$].

Even the non-relativistic limit is inconsistent.



Christian Møller

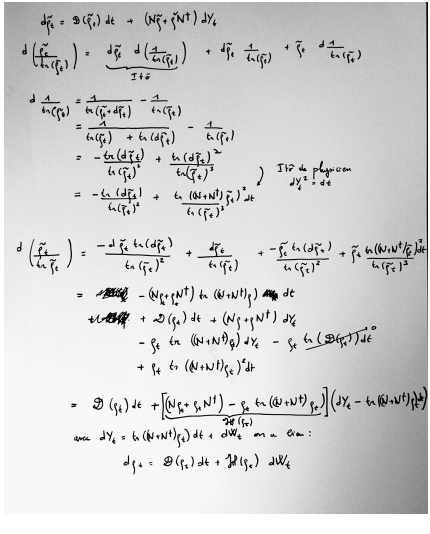


Leon Rosenfeld

Genesis at ENS - 2014

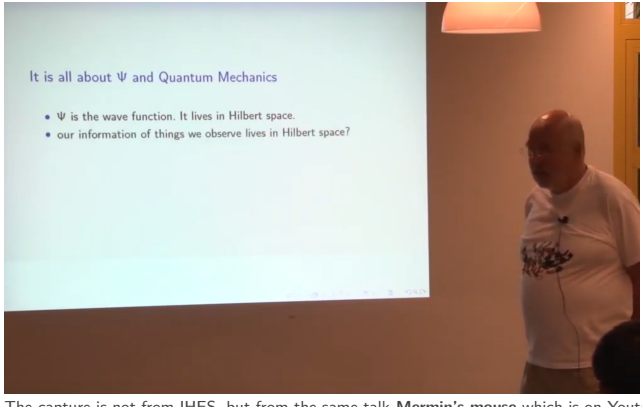
Serious continuous measurement theory – but waiting for heresy





February 2015 - IHES, Bures-sur-Yvette

At a serious semester on Quantum Mechanics organized by Jürg Fröhlich





The capture is not from IHES, but from the same talk **Mermin's mouse** which is on Youtube

After the talk and long discussions, Detlef invited me to the next foundations conference, the month after

Detlef Dürr

1951 - 2021 – moto: “Particles move” ; iconic paper [DGZ 1992]



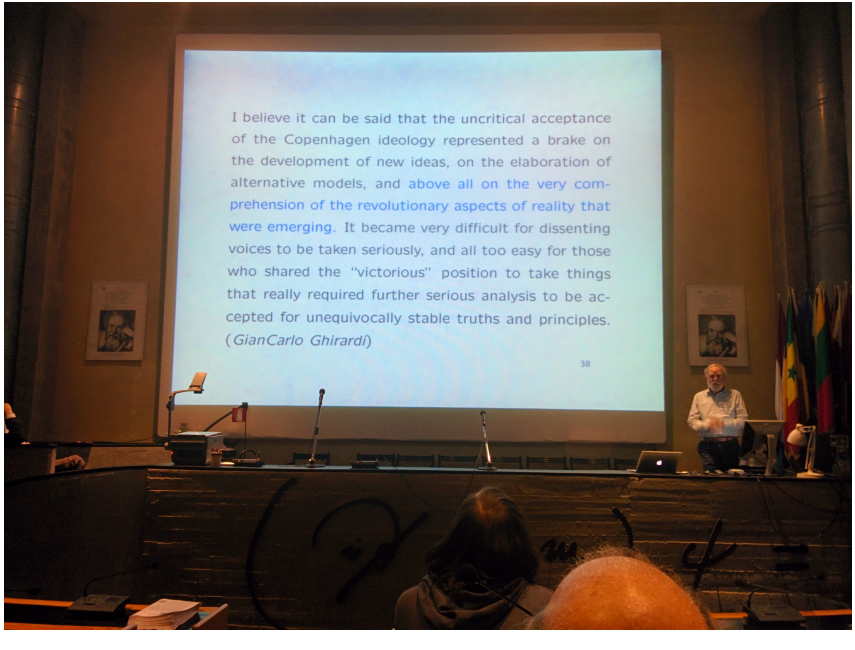


photos by Ryan Reece and Tim Maudlin

March 2015 - Erice, Sicily



March 2015 - Erice, Sicily



March 2015 - Erice, Sicily



Discussions with Lajos Diósi in Erice

Lajos Diosi was arguing with Shelly Goldstein

Which are the right local beables for QM?

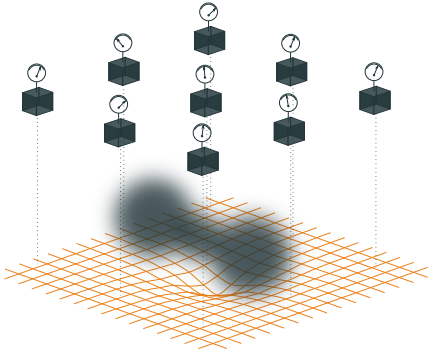
Lajos: For something to be real, it should be injectable in the Schrödinger equation as external drive without inconsistency.





From discussions to a gravity model - late 2015

Idea: The signal from continuous measurement models can be fed back without problem. Let us continuously measure mass and use the signal as source of the gravitational field to get a consistent semiclassical theory of gravity. The signal is real!



From the idea to papers 2015-2017

[Submitted on 29 Sep 2015 (v1), last revised 1 Feb 2016 (this version, v2)]

Sourcing semiclassical gravity from spontaneously localized quantum matter

Antoine Tilloy, Lajos Diósi

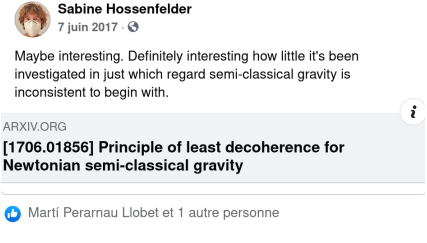
The possibility that a classical space-time and quantum matter cohabit at the deepest level, i.e. the possibility of having a fundamental and not phenomenological semiclassical gravity, is often disregarded for lack of a good candidate theory. The standard semiclassical theory suffers from fundamental inconsistencies (e.g.: Schrödinger cat sources, faster-than-light communication and violation of the Born rule) which can only be ignored in simple typical situations. We harness the power of spontaneous localization models, historically constructed to solve the measurement problem in quantum mechanics, to build a consistent theory of (stochastic) semiclassical gravity in the Newtonian limit. Our model makes quantitative and potentially testable predictions: we recover the Newtonian pair potential up to a short distance cut-off (hence we predict no 1 particle self-interaction) and uncover an additional gravitational decoherence term which depends on the specifics of the underlying spontaneous localization model considered. We hint at a possible program to go past the Newtonian limit, towards a consistent general relativistic semiclassical gravity.

[Submitted on 6 Jun 2017]

Principle of least decoherence for Newtonian semi-classical gravity

Antoine Tilloy, Lajos Diósi

Recent works have proved that semi-classical theories of gravity needed not be fundamentally inconsistent, at least in the Newtonian regime. Using the machinery of continuous measurement theory and feedback, it was shown that one could construct well behaved models of hybrid quantum-classical dynamics at the price of an imposed (non unique) decoherence structure. We introduce a principle of least decoherence (PLD) which allows to naturally single out a unique model from all the available options; up to some unspecified short distance regularization scale. Interestingly, the resulting model is found to coincide with the old --erstwhile only heuristically motivated-- proposal of Penrose and one of us for gravity-related spontaneous decoherence and collapse. Finally, this letter suggests that it is in the submillimeter behavior of gravity that new phenomena might be found.



Convincing philosophers in the Black Forest 2017



Main question: can you make your model simpler to see what is going on?

GRW toy model

- Make it easy: drop continuous Ito calculus $\rightarrow$ jump processes
- Make it familiar: use the Ghirardi-Rimini-Weber model as starting point

PHYSICAL REVIEW D **97,** 021502(R) (2018)

Rapid Communications

Ghirardi-Rimini-Weber model with massive flashes

Antoine Tilloy*

Max-Planck-Institut für Quantenoptik, Hans-Kopfermann-Straße 1, 85748 Garching, Germany

(Received 13 October 2017; published 23 January 2018)

I introduce a modification of the Ghirardi-Rimini-Weber (GRW) model in which the flashes (or space-time collapse events) source a classical gravitational field. The resulting semiclassical theory of Newtonian gravity preserves the statistical interpretation of quantum states of matter in contrast with mean field approaches. It can be seen as a discrete version of recent proposals of consistent hybrid quantum classical theories. The model is in agreement with known experimental data and introduces new falsifiable predictions: (1) single particles do not self-interact, (2) the $1/r$ gravitational potential of Newtonian gravity is cut off at short ($\lesssim 10^{-7}$ m) distances, and (3) gravity makes spatial superpositions decohere at a rate inversely proportional to that coming from the vanilla GRW model. Together, the last two predictions make the model experimentally falsifiable for *all* values of its parameters.

Anil Ananthaswamy picks it

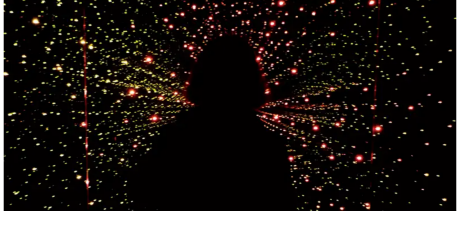
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Gravity may be created by strange flashes in the quantum realm

A model of how wave forms of quantum systems collapse reveals a way they could create gravitational fields, and perhaps even reconcile two pillars of physics

PHYSICS 20 September 2017

By Anil Ananthaswamy



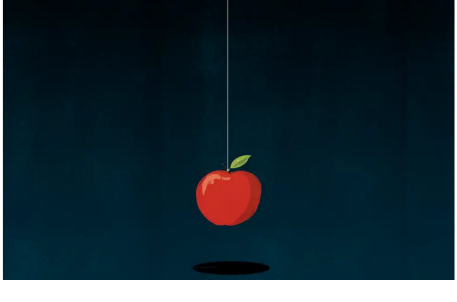
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The big fudge: Welcome to the theory of not-quite-everything

Physicists from Einstein to Hawking tried and failed to unite gravity and quantum theory. Now we have hints of a better – but not so beautiful – answer

PHYSICS 11 April 2018

By Anil Ananthaswamy



Writing popular articles

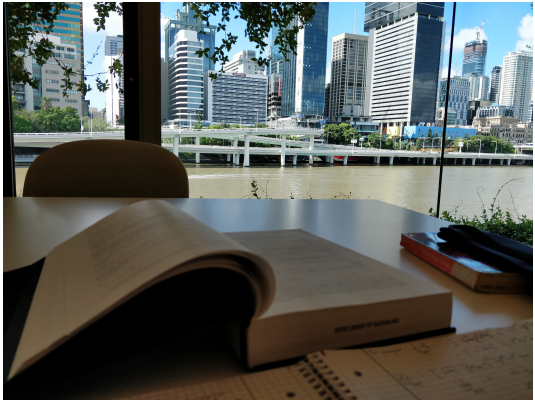




Brisbane intermezzo: to neutron stars and beyond

After my talk

- Tom Stace: your model should heat bulk matter
- Me: I know of course, $\partial_t \mathrm{tr}[H\rho_t] \geqslant 0$
- Tom Stace: what does it say for neutron stars?
- Me: No idea



Mainstream at last

After a v1 completely wrong because I do not know how to compute a polar integral (that got me rightfully destroyed) v2 in PRL

PHYSICAL REVIEW LETTERS **123,** 080402 (2019)

Neutron Star Heating Constraints on Wave-Function Collapse Models

Antoine Tilloy[1,*] and Thomas M. Stace[2]

[1]*Max-Planck-Institut für Quantenoptik, Hans-Kopfermann-Straße 1, 85748 Garching, Germany*

[2]*ARC Centre for Engineered Quantum Systems, School of Mathematics and Physics, The University of Queensland, Brisbane, Queensland 4072, Australia*

(Received 25 February 2019; published 21 August 2019)

Spontaneous wave-function collapse models, like continuous spontaneous localization, are designed to suppress macroscopic superpositions while preserving microscopic quantum phenomena. An observable consequence of collapse models is spontaneous heating of massive objects. We calculate the collapse-induced heating rate of astrophysical objects, and the corresponding equilibrium temperature. We apply these results to neutron stars, the densest phase of baryonic matter in the Universe. Stronger collapse model parameters imply greater heating, allowing us to derive competitive bounds on model parameters using neutron star observational data, and to propose speculative bounds based on the capabilities of current and future astronomical surveys.

DOI: 10.1103/PhysRevLett.123.080402

Conclusion

- **Fundamentally semiclassical gravity** is still theoretically viable. Experiments will decide.

  "I bet 99 to one that the outcome will be consistent with gravity having quantum properties."

  Carlo Rovelli

- New ideas take time to diffuse and be properly understood.
- Side projects are a good to stay motivated with moderate effect on productivity